

Introduction

My professional profile revolves around several very distinct areas that interact with each other. On one hand, there is my academic background in mathematics, which I have taken all the way to a PhD on sometimes very theoretical subjects. However, I have always had a strong attraction to all sciences. In particular, I am passionate about biology and geology (disciplines to which I was trained in CPGE BCPST notably) and I hope one day to be able to work in multidisciplinary teams. On the other hand, I started very early to learn computer code and electronics as a self-taught person, which led me to create a micro-company during my studies to work as a contractor. This mastery of computer code, more professional than academic, allows me today to develop industrial-quality software based on my mathematical work. A third important aspect of my profile has been my involvement in the cultural life of the city of Tours, as a musician in several bands but also as a scenographer and manager (volunteer) for local festivals. Through this contact with the cultural community and my mathematical and computer expertise, I was able to invent (and I still do) innovative musical instruments and musical machines.

Professional experiences

- 2025-2027 **Post-doctorate position**, *Laboratoire Jacques-Louis Lions*, Université Paris Cité
- 2021-2025 **PhD student in mathematics**, *Institut Denis Poisson*, Université de Tours
- 2021-2025 **Teacher in mathematics and computer science**, *Université de Tours*
- 2022-2025 **Teacher in mathematics and computer science**, *SUPINFO*, Tours, freelance
- 2019-2024 **Developer and electronics engineer**, *freelance*

Here is a sample of the projects I worked on:

- Development and maintaining of capture and archive software for the hospital (CHRU) of Tours, project webpage: <https://dynamise-medical.fr/capture>
- Development and maintaining of a dental 3D printing and tray thermoforming line, client website: <https://www.innower3d.com/innower-industrie/>
- Back-end development of an interface between comptability software and a government API (Urssaf), client website: <http://menadom.net/>

Publications

In peer-reviewed journals

- B. Andreianov, T. Girard, Existence of solutions for a class of one-dimensional models of pedestrian evacuations, *SIAM J. Math. Anal.* 56 (3) 3386–3411, 2024.
- P. Cardaliaguet, N. Forcadel, T. Girard, R. Monneau, Correspondance between scalar conservation law and Hamilton-Jacobi-Bellman evolution equation on a simple network in the convex case, *Discrete and Continuous Dynamical Systems*, 44(12):3920–3961, 2024.
- T. Girard, hughes2d: Approximating solutions to the Hughes pedestrian-flow model. *Journal of Open Source Software*, 11(123), 10106, 2026.

Book chapters

- D. Amadori, B. Andreianov, M. Di Francesco, S. Fagioli, T. Girard, et al., The mathematical theory of Hughes' model: a survey of results. *Crowd Dynamics* (volume 4), 2023.

Softwares

- T. Girard, Hughes2d, Github (2025). *A python package for simulating macroscopic pedestrian evacuations*, <https://github.com/TheoRGirard/hughes2d>

Preprints and other documents

- T. Girard, Discontinuités dans les équations d'Hamilton-Jacobi-Bellman et dans les lois de conservation scalaires rencontrées dans les modèles de dynamique piétonnière, thesis, Université de Tours (2025).
- J. Berry, T. Girard et F. Peru, Étude et comparaison de différents modèles de dynamique piétonnière. *MATAPLI 138* :151-153, Nov. 2025.

Academic cursus

- 2025 **Doctorat en Mathématiques**, *Université de Tours*
- 2021 **Master mathématiques approfondies**, *Université de Tours*
- 2019 **Licence de mathématiques**, *Université de Tours*
- 2015-2016 **CPGE BCPST**, *Lycée Descartes, Tours*

Distinctions

- 2026 **Prix de thèse de la ville de Tours**, *Université de Tours*

Talks

Presentations in workshops/conferences

- Jan 2024 **Conference on Numerical methods for optimal transport problems, mean field games, and multi-agent dynamics**, *Valparaíso, Chile*
- Jan 2023 **Workshop "First order problems on networks and applications"**, *Roma, Italy*

Seminars

- Mar 2026 **Seminar of the PDE team**, *Montpellier, France*
- Jan 2026 **Seminar of the PDE team**, *Besançon, France*
- Sep 2024 **Seminar of numerical analysis**, *Besançon, France*
- Dec 2022 **Seminar of the PDE team**, *Rouen, France*
- Nov 2022 **Seminar of the PDE team**, *Tours, France*
- Nov 2022 **Seminar of the PDE team**, *Bangalore, India*

Poster sessions and introductive presentations

- Nov 2023 **PhD students of the IDP day**, *Orléans, France*
- Jul 2023 **PhD students of the IDP week**, *Nouant-Le-Fuzelier, France*
- Jun 2023 **Workshop Maathraffic, Poster session**, *Tours, France*

Travels

- May 2026 **Participation to the HYP 2026 conference**, *1 week, Stuttgart, Germany*
- Mar 2026 **Meeting of the MITI project "Piecamp"**, *4 days, XLIM, Limoges, France*
- Dec 2025 **Research with F. Peru and J. Berry**, *1 week, Centrale Sup Elec, Saclay, France*

- Oct 2025 **Participation to the HypNut conference**, 1 week, Amiens, France
- Feb 2025 **Research with F. Peru and J. Berry in the context of a BOUM project**, 1 week, CIRM Marseille, France
- May 2024 **Research with Fabio Camilli**, 2 weeks, Roma, Italy
- Oct 2023 **Participation to the autumn school Ecology, epidemiology and evolution**, 1 week, CIRM Marseille, France
- Apr 2023 **Research with Regis Monneau**, 3 days, Paris, France
- Nov 2022 **Research with Shyam Goshal**, 3 weeks, Bangalore, India
- Jun 2022 **Participation to the HYP 2022 conference**, 1 week, Malaga, Spain

Organization

- 2026 **Writing and organisation of a Maths Vives project**, *MOCAIC: Modeling of Crowds: Anticipation, Information and Congestion*, With B. Andreianov, C. Appert-Rolland, J. Barré, A. Dumas, N. Igbida, V. Perrollaz, F. Silva and D. Ullmo
- 2026 **Organization of an ANR COSS meeting**, *Tours*
- 2025 **Organization of a multidisciplinary research meeting**, *backed-up by the MITI PieCamp project*, With C. Appert-Rolland, J. Berry, K. Koutsomitis, F. Peru, D. Ullmo
- 2025 **Organization of a BOUM project**, *Numerical comparison of some macroscopic models for crowd motion*, With F. Peru and J. Berry
<https://conferences.cirm-math.fr/3512.html>
- 2023 **Organization of the workshop Maathraffic**, *Tours*
<https://www.idpoisson.fr/workshop-maathraffic/>

Research project and groups

- 2022-2027 **Member of the ANR project "Control On Stratified Structures" (COSS)**
- 2025-2026 **Member of the MITI project "PieCamp"**

Teachings

Mathematics

- 2024 **Mathematical tools for biology students**, *L1 Biologie*, Université de Tours, Tours
- 2024 **Algebra III**, *L2 Mathématiques*, Université de Tours, Tours
- 2024 **Analysis III**, *L2 Mathématiques*, Université de Tours, Tours
- 2021-2025 **Linear algebra**, *First year*, Supinfo, Tours
- 2023 **Mathematical reasoning**, *L1 Mathématiques*, Université de Tours, Tours
- 2022-2023 **Geometry**, *L2 Mathématiques*, Université de Tours, Tours

Computer science

- 2025 **Supply chain and Software engineering**, *Master MECEN*, Université de Tours, Tours
- 2024 **Databases with advanced SQL**, *Second year*, Supinfo, Tours
- 2023-2025 **C programming**, *First year*, Supinfo, Tours
- 2023 **Internet of things and embedded systems**, *Second year*, Supinfo, Tours
- 2021-2023 **Python programming**, *L1 Mathématiques*, Université de Tours, Tours

Inventions et creations

Music instruments

- **Polyclav**, *A modular synth that fits in a suitcase*, (2018)
- **Soubasynth**, *An electronic sousaphone*, (2020)
- **Brassynth**, *A wind-based synth that emulates the behavior of any brass instrument*, (2021)
<https://blog.bela.io/brassynth/>
- **Sandpile oscillator**, *A machine that creates sandpiles and music*, (coming soon)

Scenography

- **Lighted signage**, *Lighted signage for the Idylle festival*, Les îlots électroniques (2025)
- **Giant teapot**, *A teapot that is 2m50 tall and opens to let a pair of eyes glow*, Les îlots électroniques (2024)
- **Metal jellyfish**, *A metal jellyfish that is embedded as the bow of a boat*, Dirty experiment (2023)
- **Infinite portholes**, *Lighted portholes that emulates infinite depth*, Les îlots électroniques (2023)
- **Giant jellyfish**, *A giant jellyfish that was floating above the main stage of the festival*, Les îlots électroniques (2023)
- **Pole platforms**, *Rolling steel plateform made to move pole dance bar across a stage*, Spicy Lotus (2021)

Computer skills

Back-end/database	javascript (nodejs), PHP, MySQL, SQLite
Software dev	python, C/C++, java, PureData (digital signal processing language)
Embedded systems	arduino, python, C#, C++, processing 5 (p5)
Web dev	javascript, html, css
Computer utils	bash, git, latex
Maths programming	python, matlab, mathematica
Technical drawing and 3D modelling	Openscad, Inkscape